

Advanced Topics in Survival Analysis

B2-Multivariate Survival Analysis.

Bayesian Survival Analysis: the frailty model.

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Outline

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 - Cox Model
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Frequentist Cox model

Let T be the random variable time to the event ϵ . The Cox model expression is

$$\lambda(t|\mathbf{X}) = \lambda_0(t) \exp(\beta_1 X_1 + \cdots + \beta_p X_p), \quad (1)$$

where $\mathbf{X} = (X_1, \dots, X_p)'$ are p covariates, and $\lambda_0(t)$ is the baseline hazard function. Moreover, remember that $\lambda_0(t)$ contains the constant term in this model and can take any shape. This is why the model is called a *semiparametric* model.

Frequentist Cox model

We know that in the frequentist approach the β 's are estimated via the partial likelihood

$$L_1(\beta_1, \dots, \beta_p) = \prod_{j=1}^r \frac{\exp(\beta' x_{(j)})}{\sum_{l \in R(t_{(j)})} \exp(\beta' x_l)}, \quad (2)$$

where $t_{(1)}, \dots, t_{(r)}$ are the r ordered times to the event of interest, $X_{(j)}$ the covariate vector for the individual failing at $t_{(j)}$, and $R(t_{(j)}) = \{i : Y_i \geq t_{(j)}\}$ is the set of individuals at risk at $t_{(j)}$. It can be proved (see the Cox's seminal paper [2]) that the β 's estimated via the maximization of (2) have great asymptotic properties. The estimate is consistent and asymptotically normal.

Bayesian Cox model

But, what happens when we use the Cox model within the Bayesian framework? Our model expression is the same

$$\lambda(t|\mathbf{X}) = \lambda_0(t) \exp(\beta_1 X_1 + \cdots + \beta_p X_p).$$

But...

- In Bayesian analysis asymptotic results do not apply.
- We should work with the complete likelihood. Makes no sense to work with partial or pseudo likelihoods.
- Semiparametric models exist, but are not easy to deal with when using more complex models (as the frailty model).

Semiparametric models

The piecewise constant hazard model is one of the most commonly used models. We do partition of the time axis, $0 < s_1 < s_2 < \dots < s_J$, with $s_J > y_i$ for all $i = 1, 2, \dots, n$. Thus, we have the J intervals $(0, s_1]$, $(s_1, s_2]$, $\dots$, $(s_{J-1}, s_J]$. In the j th interval, we assume a constant baseline hazard $\lambda_0(y) = \lambda_j$ for $y \in I_j = (s_{j-1}, s_j]$. Writing $\boldsymbol{\lambda} = (\lambda_1, \lambda_2, \dots, \lambda_J)'$, we can write the likelihood function of $(\boldsymbol{\beta}, \boldsymbol{\lambda})$ for the n subjects as follows

$$L(\boldsymbol{\beta}, \boldsymbol{\lambda} \mid D) = \prod_{i=1}^n \prod_{j=1}^J (\lambda_j \exp(\mathbf{x}'_i \boldsymbol{\beta}))^{\delta_{ij} \nu_i} \exp \left\{ -\delta_{ij} [\lambda_j (y_i - s_{j-1}) + \sum_{g=1}^{j-1} \lambda_g (s_g - s_{g-1})] \exp(\mathbf{x}'_i \boldsymbol{\beta}) \right\}, \quad (3)$$

where $\delta_{ij} = 1$ if the i th subject failed or was censored in the j th interval, and 0 otherwise.

Parametric models

In practice, when dealing with more complex models than the Cox model, we use parametric models. They are less flexible, but provide good results.

- $T \sim \text{Weibull}(\alpha, \lambda)$
- $T \sim \text{Gompertz}(\eta, b)$

When working under a parametric model the baseline hazard function is completely specified, for instance in the Weibull case $\lambda_0(t) = \lambda \alpha t^{\alpha-1}$.

Frailty model: Model expression

Let T be the time until the event of interest ϵ . Assume that the sample units are clusters (e.g. patients are clustered within hospitals, patients itself could be a cluster, etc). Suppose we have n observations, p covariates and l (possibly $l = n$) clusters. Then the model expression for the so-called *Frailty model* (or shared frailty model) is

$$\lambda(t|\mathbf{X}, \text{Cluster } k) = w_k \lambda_0(t) \exp(\beta_1 X_1 + \cdots + \beta_p X_p), \quad (4)$$

where w_k , $k = 1 \dots, l$ is the frailty term corresponding to the cluster k , $\lambda_0(t)$ is the (unspecified) baseline hazard function, $\mathbf{X} = (X_1, \dots, X_p)'$ are the p covariates of the model.

- The frailty term acts multiplicatively on the hazard.
- W is assumed to have mean 1.

Modeling the baseline hazard

There are several options

- Completely specify the hazard function assuming a parametric model.
- Using the piece-wise constant hazard function.
- Applying a diffuse gamma process prior to the baseline cumulative hazard.

See more information about the last two in [3].

Modeling the frailties

The frailty model is completely specified by taking a distribution of the frailty W across the clusters. The most common is the gamma distribution.

We assume that $W \sim \text{Gamma}(\psi, \psi)$, so $E(W) = 1$, $\text{Var}(W) = 1/\psi$.

In addition, by using this approach we only have one parameter κ to estimate. The gamma frailty offers a simple characterization of within-cluster dependence in terms of Kendall's tau being $1/(1 + 2\psi)$.

Kidney data

The reference used for this case study section is [1]. The software used is R and **Stan**¹. This software computes the posterior distributions by means of the MCMC method, specifically using the Hamiltonian Monte Carlo (HMC) method.

In this case study we use the **kidney infection data set**, available in the *frailtyHL* package in R. This dataset consists of times to the first and second recurrences of infection in 38 kidney patients using a portable dialysis machine. Infections can occur at the location of insertion of the catheter. The catheter is later removed if infection occurs and can be removed for other reasons, in which case the observation is censored.

¹See the Stan website on <https://mc-stan.org/>

Kidney data

In the kidney data set we have the following variables, each one repeated twice for each patient:

- `id`: Unique patient id.
- `time`: Time in (days) from insertion of the catheter to infection in kidney patients using portable dialysis machine.
- `status`: Censoring indicator (1 if patient is uncensored and 0 otherwise).
- `age`: Age (in years) of each patient.
- `sex`: Sex of each patient (1 if male and 2 if female).
- `disease`: Disease type (GN; AN; PKD; Other).
- `frail`: Frailty estimate from original paper.
- GN: Indicator for disease type GN.
- AN: Indicator for disease type AN.
- PKD: indicator for disease type PKD.

Kdney data

id	time	status	age	sex	disease	frail	GN	AN	PKI
1	8	1	28	1	Other	2.3	0	0	0
1	16	1	28	1	Other	2.3	0	0	0
2	23	1	48	2	GN	1.9	1	0	0
2	13	0	48	2	GN	1.9	1	0	0
3	22	1	32	1	Other	1.2	0	0	0
3	28	1	32	1	Other	1.2	0	0	0

Table: First observations of the Kidney data set.

We will be modeling using the *clock-reset* approach. It is, we are interested in the **gap time**.

Frailty model

In this case study we will model the time (in days) from insertion of a catheter in patient i th to infection by means of the frailty model:

$$\lambda_i(t|\lambda_0, \beta, w_i) = w_i \lambda_0(t) \exp(\beta_2 X_1), \quad i = 1, \dots, n \quad (5)$$

where $X_1 = 1$ if the patient is a female, and 0 if is a male, w_i is the frailty term, β_2 is the regression coefficient for the sex covariate, and n is the number of patients.

Frequentist frailty model

Call:

```
coxph(formula = kidnsurv ~ sex + frailty(id, "gaussian"), data = kidney)
```

n= 76, number of events= 58

	coef	se(coef)	se2	Chisq	DF	p
sex	-1.38	0.4327	0.3144	10.16	1.00	0.0014
frailty(id, "gaussian")				25.01	14.16	0.0370

	exp(coef)	exp(-coef)	lower .95	upper .95
sex	0.2517	3.973	0.1078	0.5878

Iterations: 6 outer, 38 Newton-Raphson

Variance of random effect= 0.5021234

Degrees of freedom for terms= 0.5 14.2

Concordance= 0.82 (se = 0.031)

Likelihood ratio test= 45.04 on 14.69 df, p=6e-05

Figure: Summary of the frequentist frailty model.

Bayesian frailty model

In order to work under the Bayesian approach we assume the Weibull parametric model, $T \sim \text{Weibull}(\alpha, \lambda)$. And then, $\lambda_0(t) = \lambda \alpha t^{\alpha-1}$. We use the following parametrization $\lambda = \exp(\beta_1)$. Hence, substituting the baseline hazard into the frailty model expression we obtain

$$\lambda_i(t|\lambda_0, \boldsymbol{\beta}, w_i) = w_i \alpha t^{\alpha-1} \exp(\beta_1 + \beta_2 X_1) \quad i = 1, \dots, n.$$

The chosen parametrization is convenient because now we treat β_1 as an intercept.

Bayesian frailty model

Now, in order to define our Bayesian model, we choose the prior distributions of all the parameters involved in our model:

- $w_i \sim \text{Gamma}(\psi, \psi)$
- $\psi \sim \text{Gamma}(0.01, 0.01)$
- $\alpha \sim \text{Unif}(0, 10)$
- $\beta_1, \beta_2 \sim N(0, 1000)$

We have chosen non-informative priors.

Hierarchical structure

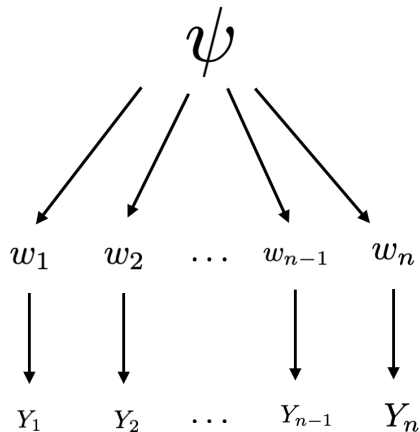


Figure: Hierarchical structure of the model.

Stan model and R libraries

```

1
2 - data {
3   int<lower=0> n;
4   int<lower=0> J;
5   int<lower=0> p;
6   matrix[n,J] time;
7   matrix[n,p] X;
8   matrix[n,J] is_censored;
9 - }
10
11 - parameters {
12   vector[p] beta;
13   vector<lower=0>[n] w;
14   real<lower=0> psi;
15   real<lower=0> alpha;
16 - }
17

```

(a)

```

18
19 - model {
20   for (i in 1:n) {
21     for (j in 1:J) {
22       if(is_censored[i,j]==1){
23         target += weibull_lccdf(time[i,j]| alpha, (w[i]*exp(beta[1]*X[i,1] + beta[2]*X[i,2]))^(-1/alpha));
24       } else{
25         time[i,j] ~ weibull(alpha,(w[i]*exp(beta[1]*X[i,1] + beta[2]*X[i,2]))^(-1/alpha));
26       }
27     }
28   }
29   alpha ~ uniform(0, 10); // Shape parameter prior
30   beta ~ normal(0, 1000); // Scale parameter prior
31   w ~ gamma(psi, psi); // Frailty parameter prior
32   psi ~ gamma(0.01,0.01);
33 - }
34
35 - generated quantities{
36   real<lower=0> lambda;
37   real<lower=0> hr;
38   lambda = exp(beta[1]);
39   hr = exp(beta[2]);
40 - }
41

```

(b)

Figure: Stan model code.

The package `rstan` is needed. For some of the plots `bayesplot` package.

Checking convergence

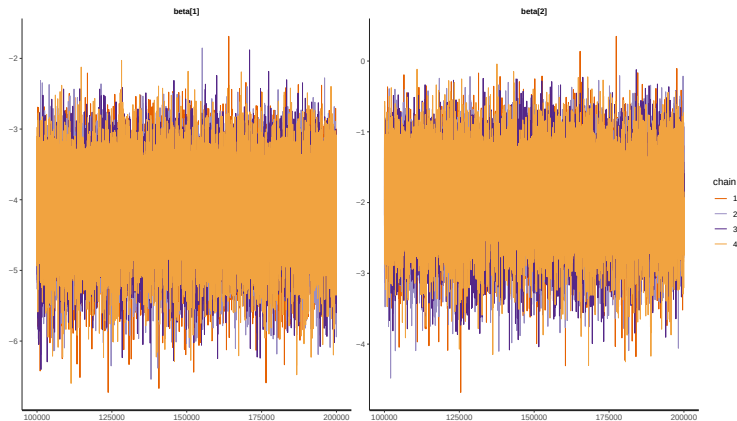


Figure: Markov chains trace plots for the simulations of the parameters β_1, β_2 .

Results of the Bayesian model

Parameter	Mean	SD	2.5%	50%	97.5%	n_{eff}	$\hat{R}$
β_1	-4.12	0.59	-5.35	-4.09	-3.02	38838	1
β_2	-1.90	0.56	-3.05	-1.88	-0.87	31455	1
α	1.23	0.16	0.93	1.22	1.57	29240	1
ψ	2.75	5.86	0.79	1.89	8.54	7514	1
λ	0.02	0.01	0.00	0.02	0.05	39166	1

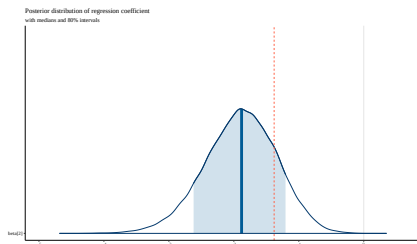
Table: Summary statistics of the posterior distributions. In addition the number of effective parameters and the $\hat{R}$ quantities.

Remember that in the frequentist model we have obtained $\hat{\beta}_2 = -1.38$.

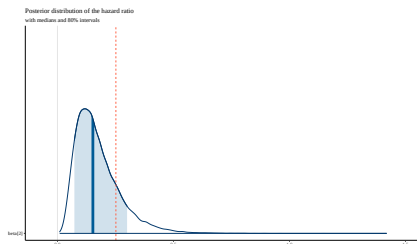
Results

The results obtained fitting a frequentist frailty model are the **point estimates** of the parameters and its standard error. However, under the Bayesian approach, what we have obtained is the **posterior distribution** of this parameters. In our case, we have a total amount of 40000 **simulations** of all the parameters involved in the model.

Results



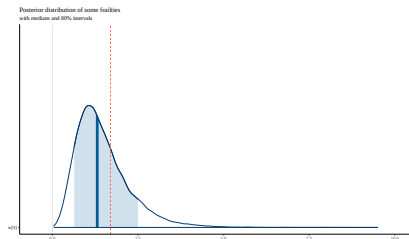
(a) Posterior distribution of β_2



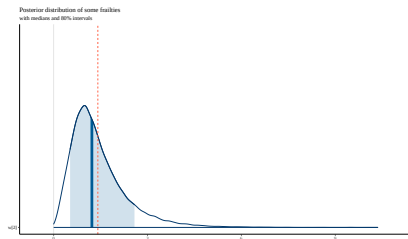
(b) Posterior distribution of $\exp(\beta_2)$

Figure: In the left, posterior distribution of β_2 , in the red dashed line the point estimate of the frequentist frailty model. In the right, the same for $\exp(\beta_2)$, the hazard ratio.

Results



(a) Posterior distribution of w_1



(b) Posterior distribution of w_2

Figure: In the left, the posterior distribution of the frailty term for the first individual, where the red dashed line is the frequentist estimation. In the right, the same but for the second individual.

Results

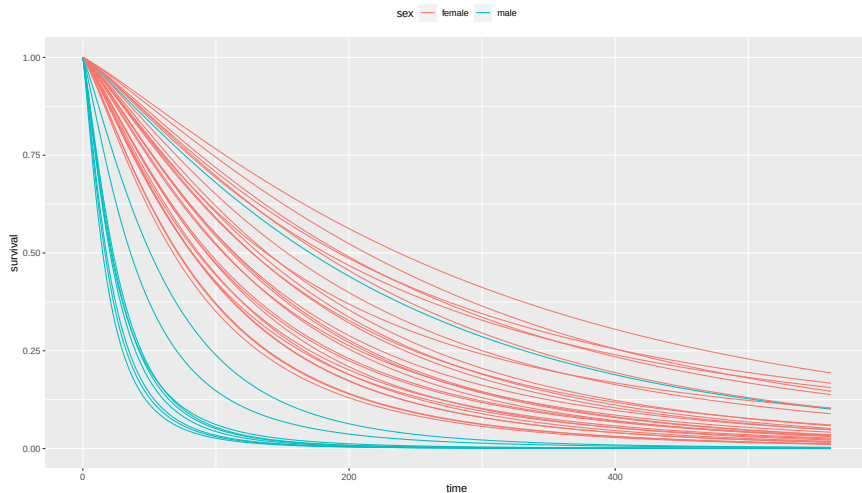
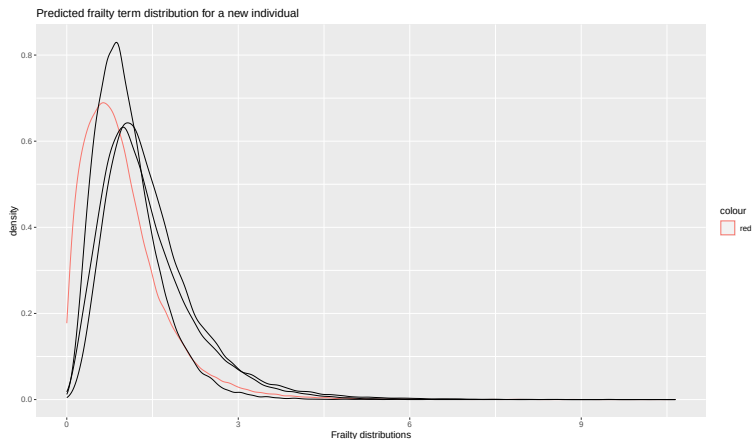


Figure: Posterior mean of the individual survival function.

Results

Exploiting the **hierarchical structure** we can compute the frailty term distribution prediction for a new individual, using

$$\pi(\tilde{w}) = \int \pi(\tilde{w}|\psi)\pi(\psi|y)\partial\psi:$$



Further material on my GitHub

You can find the whole code used, and a memory with the topics presented and more theoretical details in my GitHub:

https:

`//github.com/ArnauGF/MSc-UPC/tree/bayesian_survival`

References



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Journal of the Royal Statistical Society: Series B (Methodological), 34(2):187–202, 1972.



John P Klein, Hans C Van Houwelingen, Joseph George Ibrahim, and Thomas H Scheike.

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